L. While the precise benchmark accuracies vary depending on the setup, we find that the general trends of overfitting hold consistently across our ablations. We will release the full evaluation code for transparency.

In addition to evaluating widely known models, we additionally evaluate several lesser known models that sit near the top of the OpenLLMLeaderboard and discover evidence of Goodhart's law: many of these models perform substantially worse on GSM1k, suggesting that they are primarily gaming the GSM8k benchmark rather than improving model reasoning capabilities. The full set of results, including the performance table for all models, can be found in Appendix F. For fair comparison, we partition the models by performance on GSM8k and compare them to other models which perform similarly (Figures 4, 11, 12).

5 Analysis

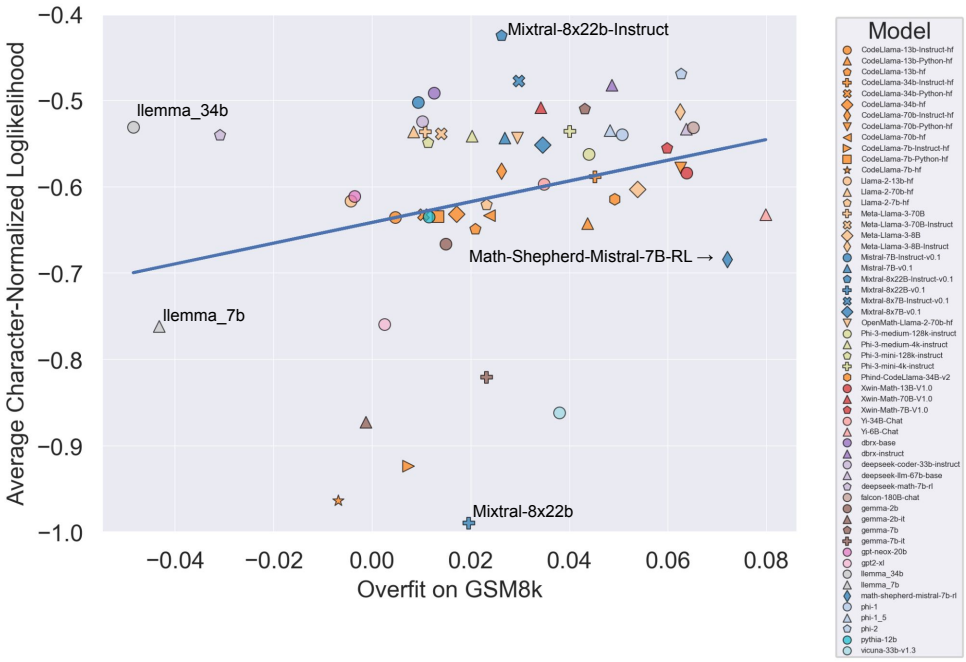

Figure 5: Comparison between overfit on GSM8k (x-axis) and average sequence-level log-likelihood on the GSM8k test set (y-axis). We find that there is a correlation between overfit on GSM8k and sequence-level log-likelihood, suggesting that, in general, models that have a high overfit generally have a higher probability of generating the test set. This suggests that some of the GSM8k test set may have leaked into the model training data. The line of best fit is in blue. Additionally, we highlight 5 "outlier" models which we discuss further with Lesson 4.

The interpretation of evaluation results, like the interpretations of dreams, is often a very subjective endeavor. While we report our objective results in Section 4 and Appendix F, here we describe four major takeaways from interpreting the results in a more subjective manner.

5.1 Lesson 1: Some Model Families are Systematically Overfit

While it is difficult to draw conclusions from singular data points or model releases, examining a family of models and observing a pattern of overfitting enables us to make more definitive statements. Several families of models, including the Phi and Mistral families of models, show systematic tendencies to perform stronger on GSM8k compared to GSM1k for almost every release and scale of models. Other model families, such as Yi, Xwin, Gemma and CodeLlama also show this pattern to a lesser extent.

5.2 Lesson 2: Other Models, Especially Frontier Models, Show No Signs of Overfitting

Nevertheless, we find that many models, through all regions of performance, show minimal signs of being overfit. In particular, we find that all frontier or close-to-frontier models (including the proprietary Mistral Large) appear to perform similarly on both GSM8k and GSM1k. We posit two potential hypotheses for this: 1) frontier models have sufficiently advanced reasoning capability so that they can generalize to new problems even if they have already seen GSM8k problems in their training set, 2) frontier model builders may be more careful about data contamination.

While it is impossible to know for certain without looking at the training set for each model, one piece of evidence in favor of the former is that Mistral Large is the *only* model in the Mistral family to show no signs of overfitting. Since the hypothesis that Mistral took unique care in ensuring that only their largest model was free from data contamination seems unlikely, we lean instead towards the hypothesis that sufficiently strong LLMs also learn elementary reasoning ability during training. If a model learns strong enough reasoning capabilities to solve problems of a given difficulty, it will be able to generalize to new problems even if GSM8k has appeared in its training set.

5.3 Lesson 3: Overfit Models Are Still Capable of Reasoning

One worry about model overfitting is that models are incapable of reasoning and are only memorizing answers seen in the training data. Our results do not support this conjecture. The fact that a model is overfit does not mean that it is poor at reasoning, merely that it is not as good as the benchmarks might indicate it to be. In fact, we find that many of the most overfit models are still capable of reasoning and solving novel problems. For example, while Phi-2 has a 6% drop in accuracy between GSM8k and GSM1k, we find that it is still able to correctly solve over half of GSM1k problems – which are certain to not have appeared in its training distribution. This performance is similar to that of much larger models such as Llama2-70B, which contains over 25x as many parameters. Similarly, Mistral models remain some of the strongest open source models, even accounting for their overfitting. This provides additional evidence for our lesson that sufficiently strong models learn elementary reasoning, even if benchmark data accidentally leaked into the training distribution, as is likely to be the case for the most overfit models.

5.4 Lesson 4: Data Contamination Is Likely Not The Full Explanation for Overfitting

A priori, a natural hypothesis is that the primary cause for overfitting is data contamination, e.g. that the test set was leaked in the pre-training or instruction fine-tuning part of the model creation. Previous work has suggested that models put higher log-likelihoods on data that they have seen during training (Carlini et al. [2023]). We test the hypothesis that data contamination is the cause of overfitting by measuring a model's probability of generating an example from the GSM8k test set and comparing it to how overfit it is on GSM8k vs GSM1k, using the assumption that a model's probability of generating the GSM8k test set is a proxy for whether the sequence is likely to have appeared in the training set. We normalize by c, the number of characters in the sequence, to make the log-likelihood calculations comparable between sequences and models with different tokenizers. Formally, we have:

$$\frac{1}{c}\sum_{i}\log p(x_i|x_{<i}) \tag{1}$$

with c being the number of characters in the sequence. Figure 5 shows the result of this plot against the gap between GSM8k and GSM1k performance. We indeed find a positive relationship between the two values. We observe a Spearman's rank correlation of 0.36 between the per-character log-likelihood of generating GSM8k and the performance gap between GSM8k and GSM1k ($p = 0.03$), and the relationship suggests that every percentage point difference in GSM8k and GSM1k performance is associated with an increase of 1.2×10^{-2} in the per-character log-likelihood. This result suggests that some of the reason for overfitting is due to partial memorization of the test set. For completeness, we also report the standard Pearson $r^2 = 0.26$ and the Kendall's τ of 0.29, but note that Pearson r^2 is not the ideal metric due to the curve-of-best-fit not appearing linear.

Nevertheless, data contamination is likely not the full story. We observe this via the presence of several outliers, which cause the $r^2 = 0.36$ value to be relatively low. Examining these outliers carefully reveals that the model with the lowest per-character log-likelihood (Mixtral-8x22b) and the model with the highest per-character log-likelihood (Mixtral-8x22b-Instruct) are not only variations of

the same model, but also have similar levels of overfit (Jiang et al. [2024]). Perhaps more intriguingly, one the most overfit models we discovered (Math-Shepherd-Mistral-7B-RL (Yu et al. [2023])) had a relatively low per-character log-likelihood. Math Shepherd trains a reward model on process level data using synthetic data. As such, we hypothesize that the reward modelling process may have leaked information about the correct reasoning chains for GSM8k even if the problems themselves did not ever appear in the dataset. Finally, we observe that the Llema models (Azerbayev et al. [2024]) have both high log-likelihoods and minimal overfit. These models are open-sourced alongside their training data, and the authors report finding a very small number of GSM8k examples in the training corpus. Nevertheless, they also find (and our study supports) that these few instances do not lead to overfitting. The existence of these outliers suggests that overfitting on GSM8k is not purely due to data contamination, but rather may be through other indirect means, such as model builders collecting data similar in nature to benchmarks as training data or selecting final model checkpoints based on performance on benchmarks, even if the model itself may have not seen the GSM8k dataset at any point via training. Conversely, the reverse is also true: small amounts of data contamination do not necessarily lead to overfitting.

6 Discussion

We create GSM1k, a novel dataset designed to measure LLM overfitting on GSM8k. When benchmarking leading open- and closed-source models, we find substantial evidence that many models have been contaminated by benchmark data, with models showing performance drops of up to 8% accuracy. Additionally, we find that several model families show consistent overfitting across almost all model sizes and versions. An extended analysis reveals a positive relationship between a model's likelihood of generating data points in GSM8k and its performance difference between GSM8k and GSM1k, suggesting evidence of data contamination as one of the underlying causes. Nevertheless, we find that frontier models exhibit little to no evidence of overfitting and that many models, even the most heavily overfit families, show strong signs of generalizable mathematical reasoning.

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